Ohio University, Department of Physics and Astronomy

6021, Homework 2, due Mar 4, 2024

1. Calculate the scattering cross section in the Born approximation for:

(i) the square well of of depth V_0 and radius a.

(ii) the Gaussian potential $V = V_0\ exp(-r^2/a^2)$

For each potential, what is the asymptotic form for high energies of the total cross section: $\int d\Omega\ \ d\sigma/d\Omega$?

2. Protons with energy 1.0MeV scatter off a fixed spherical potential whose range is known to be $a = 10^{-13}$ cm. The total elastic scattering cross section σ and the differential cross section $d\sigma/d\Omega$ are measured to a precision of about one percent.

(a) Assuming that the energy is not near any resonance of the potential, *show* that the only phase shifts that can be reliably determined from the data are δ_0 and δ_1.

(b) Given the experimental value $\sigma = (10.5 \pm 0.1) \times 10^{-26} cm^2$ and the values for $d\sigma/d\Omega$ listed in the table (one percent accuracy), *determine* δ_0 and δ_1. (Do not worry about the errors in your values for δ_0 or δ_1.

θ (deg)	$d\sigma/d\Omega$ (10^{-26} cm^2)
0	1.232
45	1.103
90	0.823
135	0.585
180	0.499

3. Analysis of the scattering of particles of mass m and energy E from a fixed scattering center with characteristic length a finds the phase shifts:

$$\delta_l = sin^{-1}[(iak)^l/\sqrt{(2l+1)l!}]$$

a) Write a closed form expression for the total cross section as a function of incident energy E.

b) For what range of E does s-wave scattering provide a good estimate of σ?

4. Consider a probability distribution $\rho(x)$ defined on the whole real line. Suppose that we know only the first three power moments of ρ:

$$\mu_n = \int_{-\infty}^{\infty} dx x^n \rho(x)$$

for $n = 0, 1, 2$.

a) Write down the entropy functional with three Lagrange multipliers to handle the constraints (moment conditions).

b) Find the maximum entropy solution to part a) using the Euler-Lagrange equation as discussed in class.

c) Give an analytic expression for the maximum-entropy density $\rho_M(x)$ in terms of the known moments.